

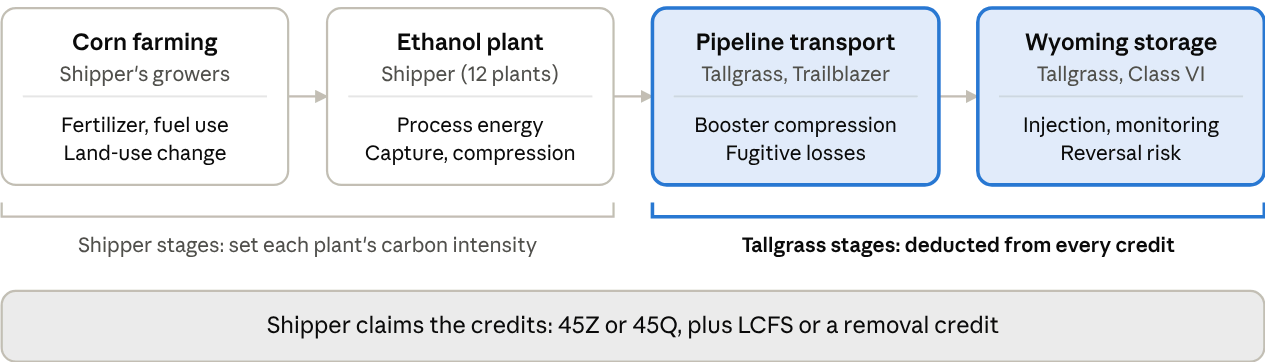
Applying the LCA Framework at Tallgrass

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Tallgrass should treat LCA as a product it sells, not a compliance chore. The storage-stage data it already collects can raise credit yield for all 12 shippers and screen every new CO2 source before it signs.

Where Tallgrass sits in the LCA

Tallgrass controls the two stages buyers audit most closely



Trailblazer LCA chain · 4 stages, 2 owners

Shippers set most of a tonne's carbon intensity, but Tallgrass produces the storage proof that every credit relies on. Its own emissions are subtracted from every shipper's result.

Three plays

Play	What Tallgrass does	Why it pays	Who benefits
1. Own the storage-stage LCA	Publish one audited data package: metered injection, booster compression power, fugitive losses, Class VI monitoring, mass balance	Every shipper needs it for CARB pathways, Puro credits and buyer diligence. Today each would assemble it separately.	All 12 shippers; Tallgrass earns a fee or revenue share

Play	What Tallgrass does	Why it pays	Who benefits
2. Grade tonnes by full LCA	Score each shipper's tonnes on net removal per captured tonne: corn sourcing, plant energy, iLUC treatment	Strict buyers (Microsoft, Frontier) pay more for tonnes that survive the CarbonPlan critique. Others go to spot.	Shippers with low-CI corn; buyers get a clear quality signal
3. Screen every new CO2 source	Run the four-phase LCA before signing any new shipper or rail-delivered CO2	Shows which credits a source can earn, and whether it dilutes the removal story for existing shippers	Tallgrass's commercial team; protects the hub's reputation

Play 1 is the quickest win. ADM Columbus already relies on Tallgrass data to certify more than 800 kt/yr with Puro. The other 11 shippers are smaller and less likely to build this themselves.

Lowering Tallgrass's own emissions adds credits directly. Puro deducts transport and injection emissions from each shipper's credits. Booster compression run on a coal-heavy grid could cost a few percent of every tonne (our estimate, to check against metered power). Cleaner power or efficiency gains turn straight into more credits for shippers.

Test case: coal capture instead of corn

Verdict: not instead of corn, and only possibly alongside it. Coal CO2 earns one credit (\$85/t under 45Q), while corn CO2 earns two or three layers. Coal's case rests on filling pipeline capacity, not on the credits.

Running the four phases on a coal tonne:

Factor	Corn ethanol tonne (today)	Coal power plant tonne
CO2 purity at source	About 99% from fermentation	About 12–15% in flue gas (approximate)
Capture cost	Low \$40s/t (Red Trail Energy)	Roughly \$60–100+/t, plus an energy penalty of about 20–30% of plant output (approximate, industry estimates)

Factor	Corn ethanol tonne (today)	Coal power plant tonne
Federal credit	45Z, about \$195–260/t through 2029, then 45Q at \$85	45Q at \$85/t only, for 12 years
Clean fuel credits (LCFS, CFP)	Yes, with a certified CCS pathway	No, electricity is not a transport fuel
Removal credits	Yes, under Puro; contested at Microsoft	No. Fossil CO ₂ stored is avoided emissions, not removal, so no removal buyer takes it
LCA result	Partly net-negative, depending on how land use is treated	Still a net emitter, from uncaptured CO ₂ (about 10%), mine methane and extra coal burned for the energy penalty
Volume	0.1–0.8 Mt/yr per plant	Several Mt/yr per plant; Dry Fork's CCS design study targets 3 Mt/yr (MTR)
Regulatory push	Market-driven	Weakening. EPA repealed the CCS-based standards for existing coal plants on Sep 14, 2026 (Federal Register). Wyoming's HB 200 study mandate survived a 2026 repeal attempt (Cowboy State Daily)

The strongest case for coal (steelman). Trailblazer has more than 10 Mtpa of capacity, and current flow is about 3 Mt (our estimate). One coal unit could fill much of that gap, and Tallgrass is paid for transport and storage, not for credits. As retirements are pushed back, the plants will be around longer.

Why it is weak today. At \$85/t, 45Q barely covers capture cost before transport and storage fees. With no federal rule requiring CCS, the plant owner has little reason to pay those fees. DOE cancelled \$49M for the Dry Fork capture pilot ([WyoFile](#)).

The LCA risk for existing shippers. Mixing fossil and biogenic CO₂ in one pipeline is allowed under Puro if Tallgrass keeps a mass balance. Premium removal buyers may still discount a hub that also stores coal CO₂. Stored pore space also has an opportunity cost if hub capacity ever becomes the constraint.

What would change the answer: a utility willing to sign a long-term take-or-pay contract, a higher 45Q rate, or a new state mandate. Ask which plant, what its retirement date is, and whether its owner has costed capture under HB 200.

First 90 days

1. **Days 1-30: map the inventory.** List every emission source Tallgrass controls, from booster stations to monitoring, with metered data. Confirm what ADM Columbus already gets from Tallgrass for its Puro audit.
2. **Days 31-60: build the shared package.** Turn that into one storage-stage LCA module that every shipper can cite for CARB pathways, Puro and buyer diligence. Price it as a fee or a revenue share.
3. **Days 61-90: score and screen.** Run net-removal-per-captured-tonne scores for two or three volunteer shippers. Use the same four phases to screen one new source, rail-delivered or coal, before any term sheet.

Questions to settle with the team:

- Under shipper contracts, who owns the removal attribute and the LCA data?
- What is the metered power use and grid source for Trailblazer's compression?
- Does Puro's corn land-use charge apply in full to ADM Columbus, or does California's iLUC accounting offset it?
- Is hub capacity likely to bind before 2040? That decides whether low-value tonnes such as coal have an opportunity cost.

Sources: Trailblazer Low-Carbon Market Strategy (Oct 2026) and the links in each section. Ranges marked approximate are industry rules of thumb, not site data.